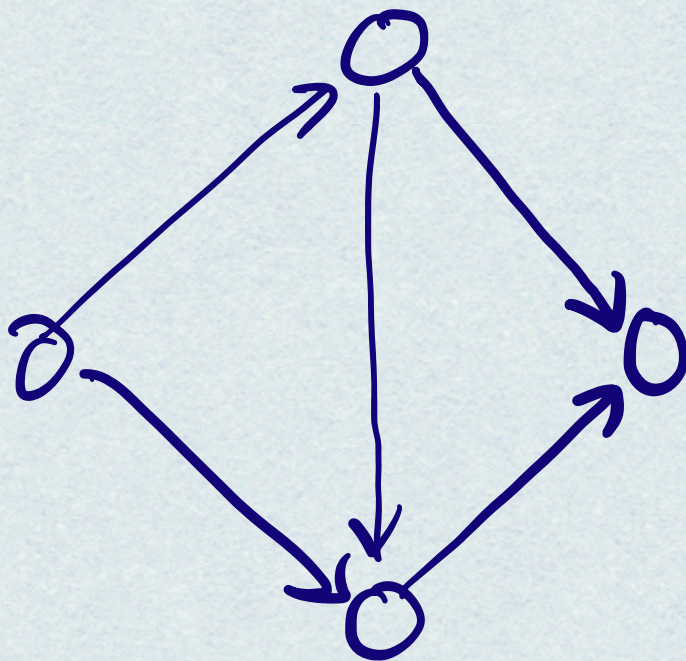


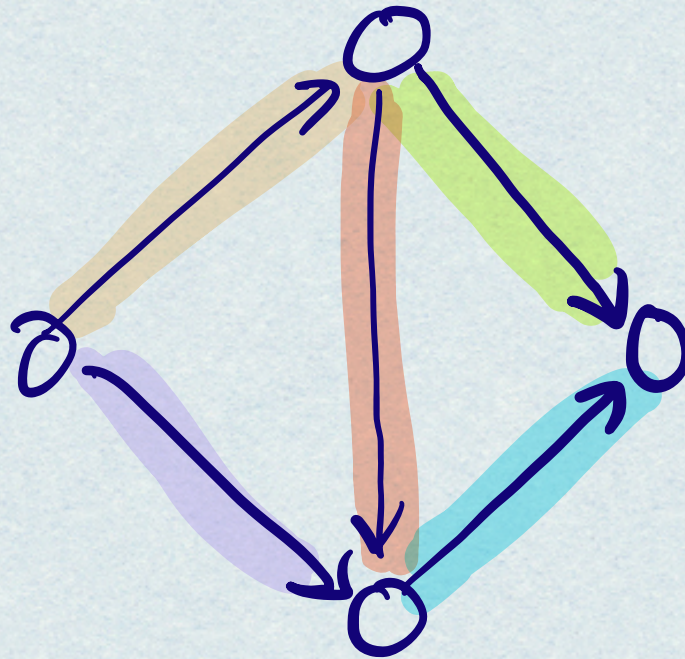
Network
Flow



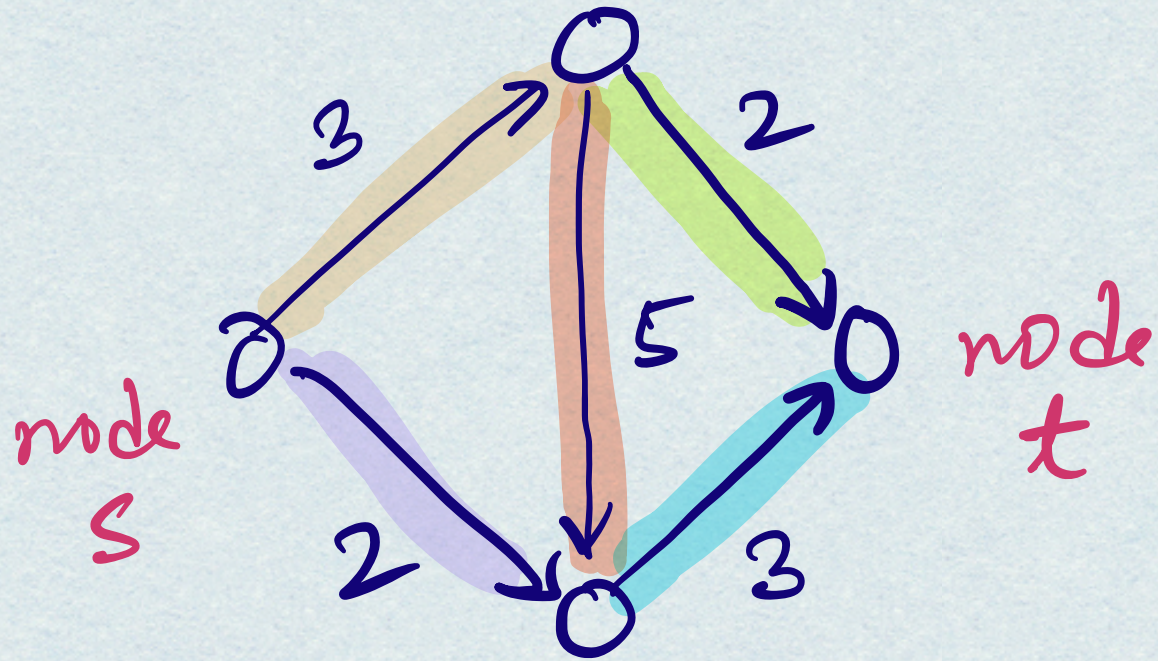
Maximum Flow problem

Say, we have a directed network.

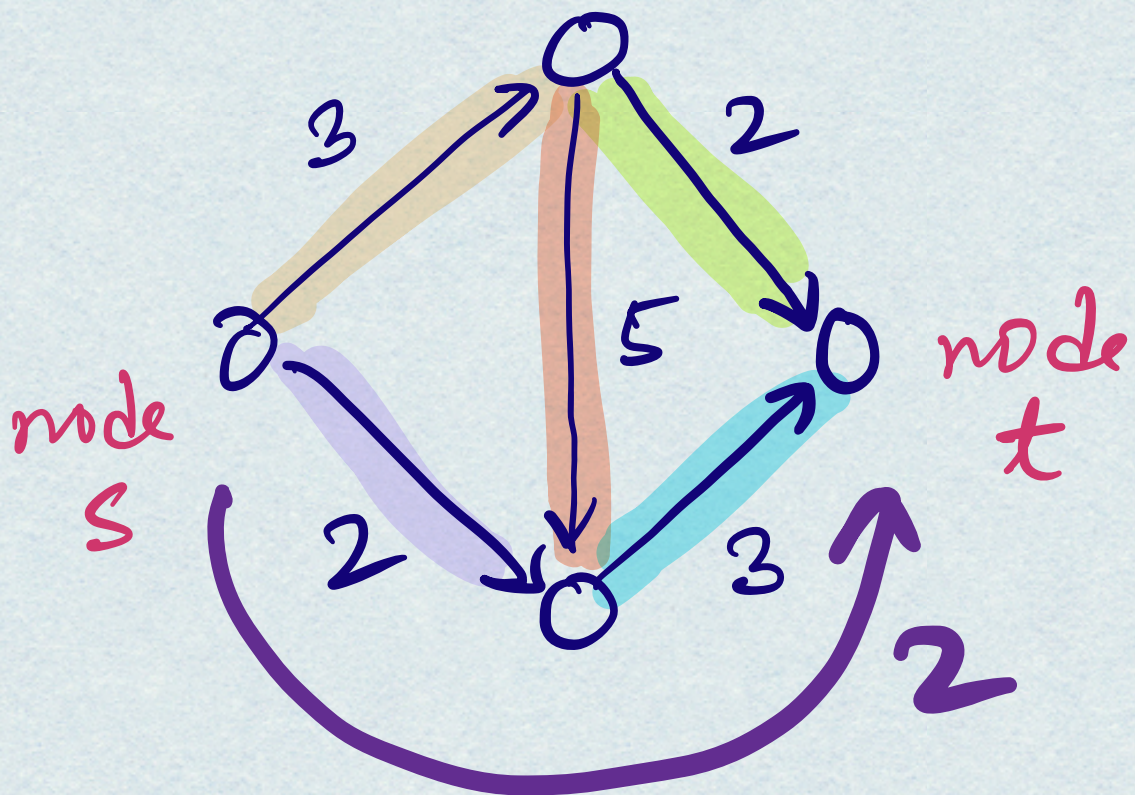
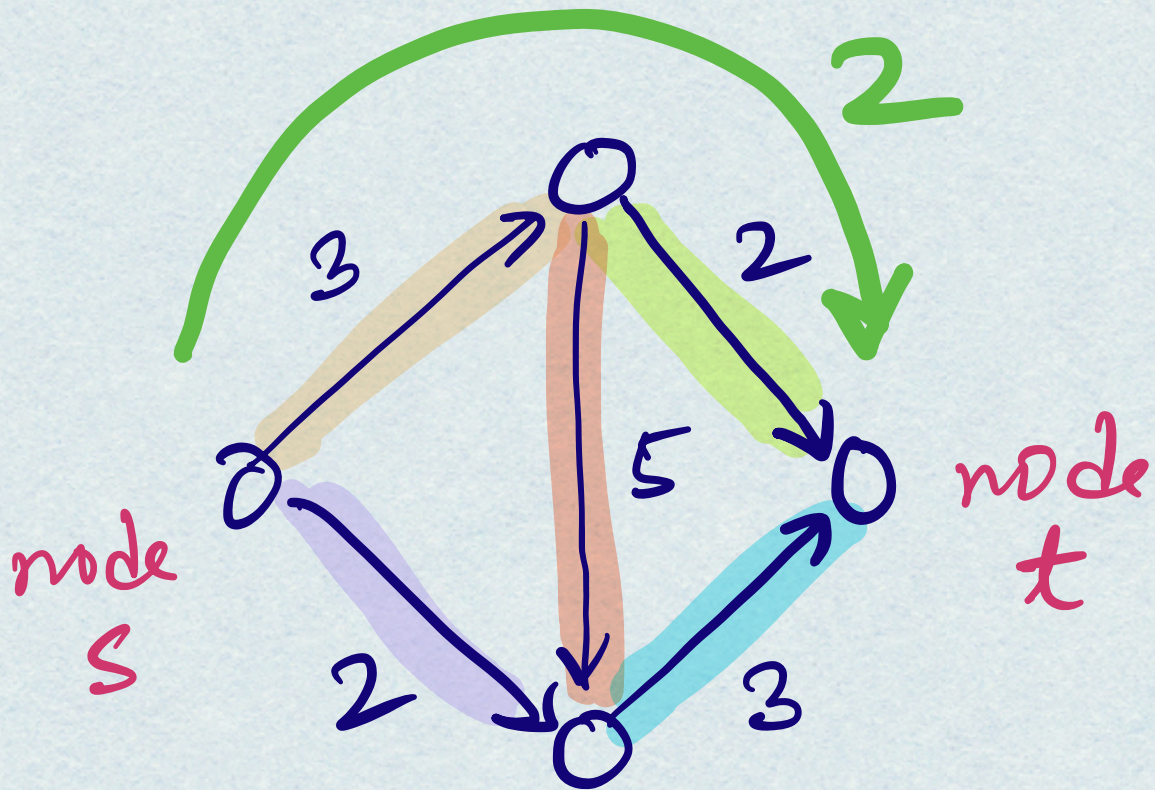


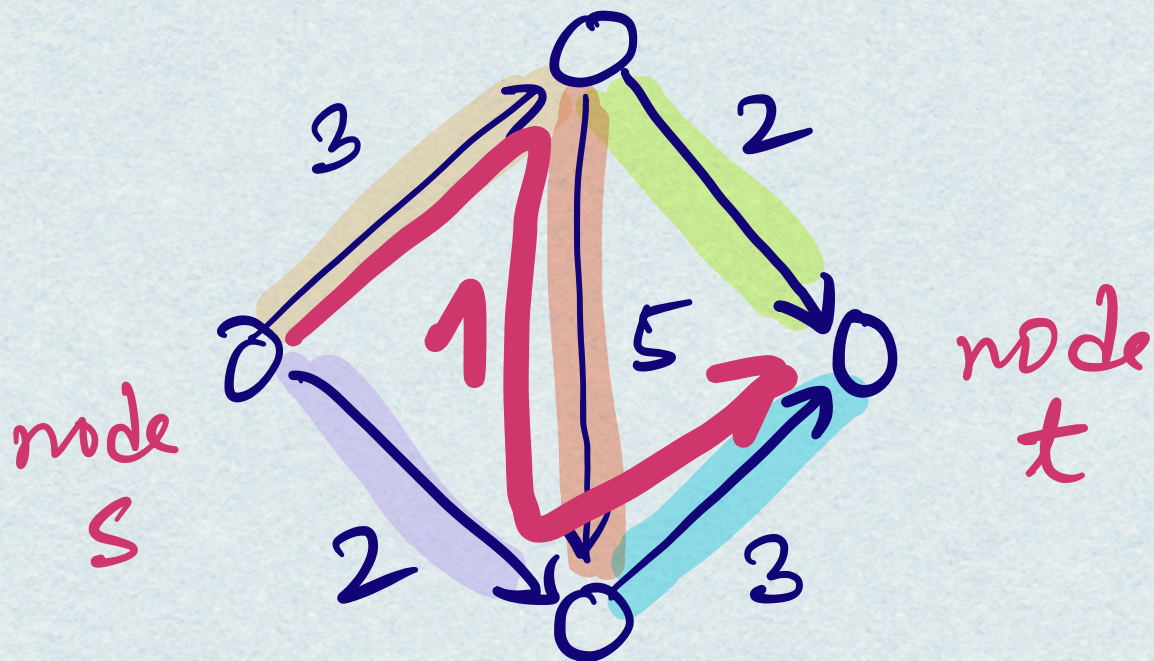


Capacity - Maximum amount of things an edge can carry.



Q. How much amount
Can be pushed
from s to t.





Total =

$$1 + 2 + 2$$

$$= \underline{5 \text{ unit}}$$

Can we do better?

— No!

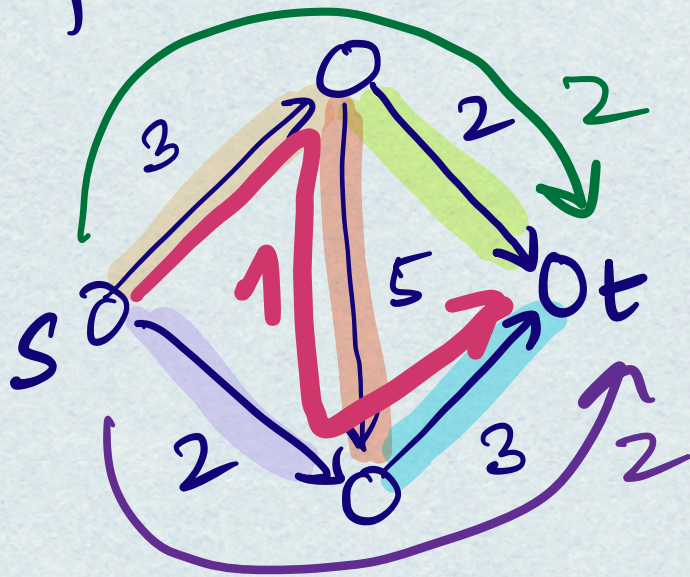
Let's define an instance
of Max-Flow.

A directed graph $G=(V,E)$

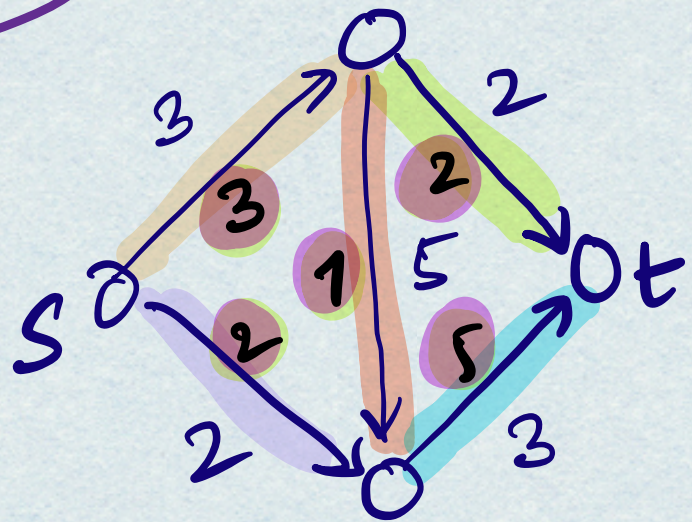
A Source vertex $s \in V$

A Sink vertex $v \in V$

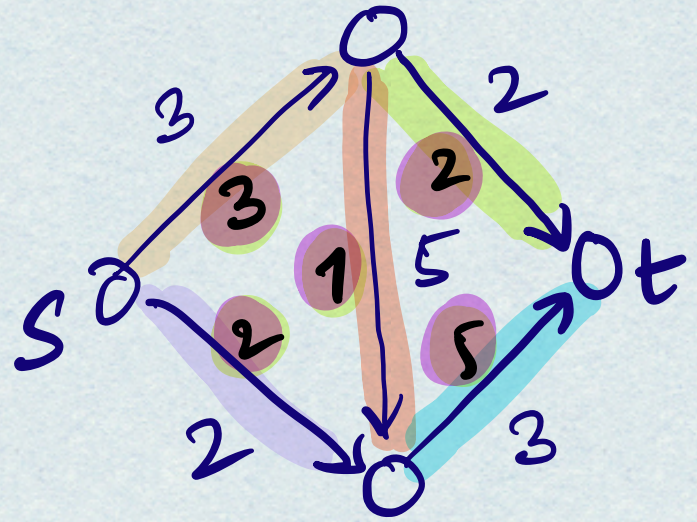
A non-negative integral capacity and u_e
for each edge $e \in E$



Superposition
of paths



Feasible
Solutions
- Flows



A non-negative
vector $\{f_e\}_{e \in E}$.

1. Capacity Constraints

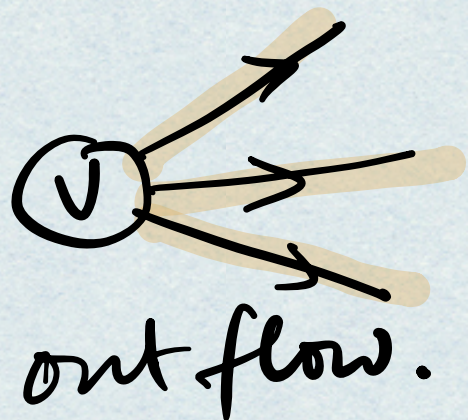
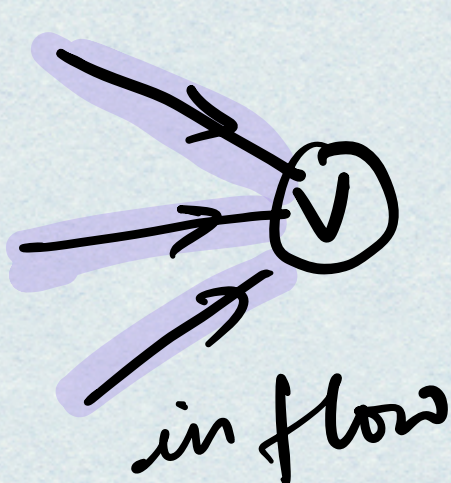
2. Conservation Constraints.

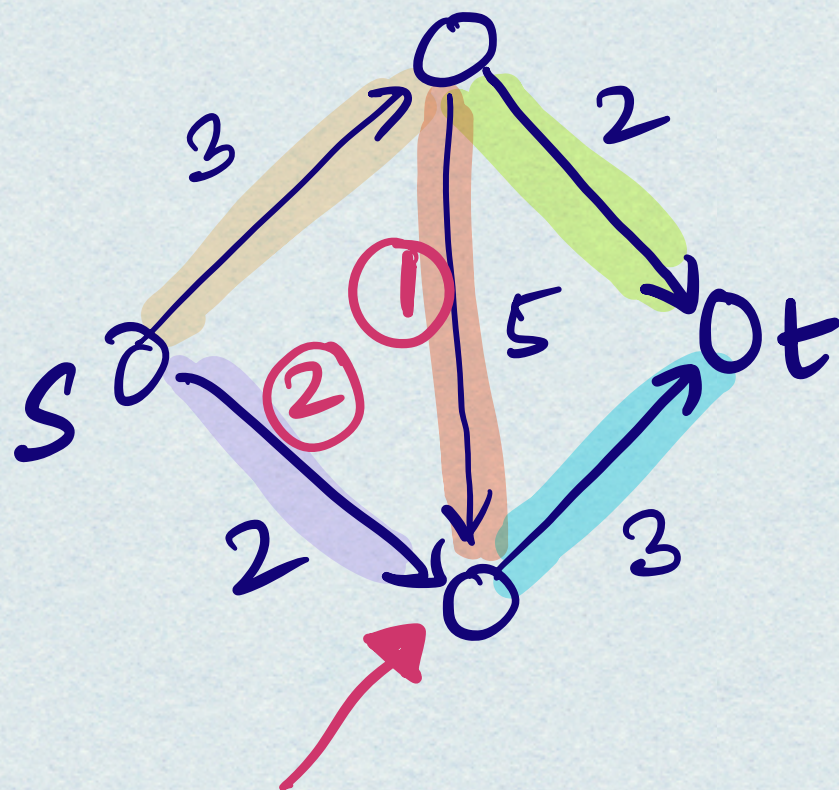
1. Capacity Constraints

$$f_e \leq u_e, \forall e \in E.$$

2. Conservation Constraints.

$$\text{flow in} = \text{flow out} \\ \forall v \neq s, t.$$



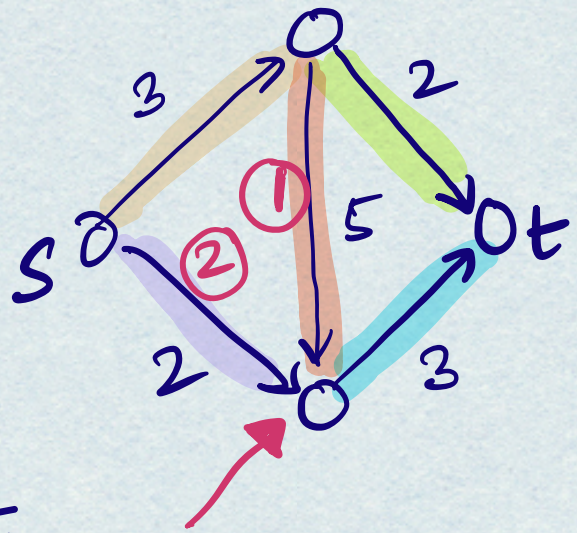


$f_e = 0$, $\forall e$ is
feasible

Claim -

Flow out
of $s =$

Flow into t



Proof:-

Define

Type I edges: (s, v) for
 $v \notin \{s, t\}$

Type-II edges:

(v, t) for $v \notin \{s, t\}$

Type-III edges:

(u, v) for $u, v \notin \{s, t\}$

$$\sum_{v \neq s, t} \text{flow-in}(v) = \quad \text{--- ①}$$
$$\sum_{\text{Type-I}} f_e + \sum_{\text{Type-III}} f_e$$

$$\sum_{v \neq s, t} \text{flow-out}(v) = - \quad (2)$$

$$\sum_{\text{Type-II}} f_e + \sum_{\text{Type-III}} f_e$$

Capacity Constraints.

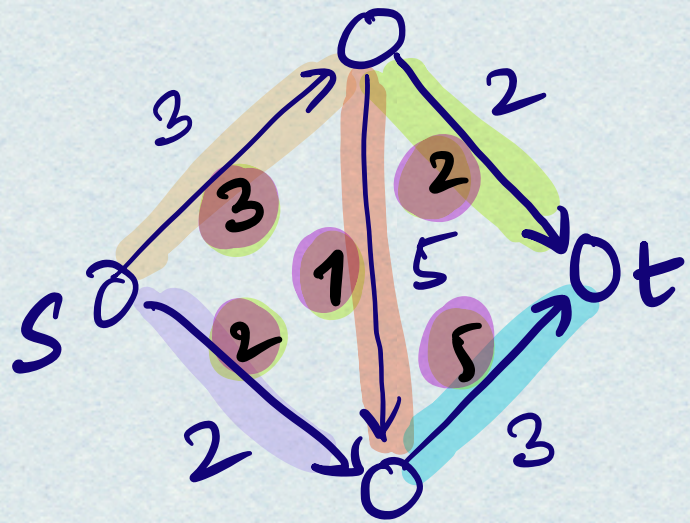
$\Rightarrow$

$$\textcircled{1} = \textcircled{2} \Rightarrow$$

flow out of s = flow to t

(Type-II) (Type-II)

Feasible
Solutions
- Flows



A non-negative
vector $\{f_e\}_{e \in E}$
Satisfying

1. Capacity Constraints
2. Conservation Constraints.

Objective:- Maximize

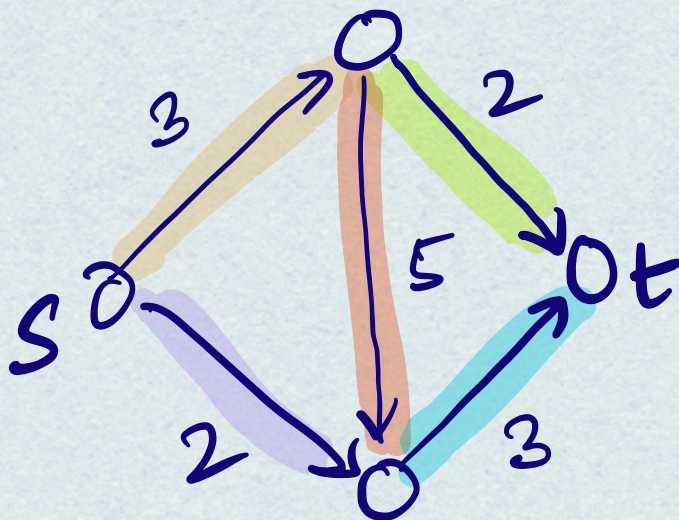
flow-time = flow going
out of s .

Assume - No edges going
into s or coming out
of t (w.l.o.g.).

Let's try Algorithms

Idea - Greedy?

Compute a sequence
of flows with progres-
sively higher values



① Initialize $f_e = 0, \forall e$.

② Repeat

- Search for an
s-t path p
s.t. $f_e < u_e \forall e \in p$

// possible in $O(|E|)$
time via BFS/DFS

- If no such path,
Return current flow
 $\{f_e\}_{e \in E}$.

- Else, let

$$\Delta = \min_{e \in P} \underbrace{(u_e - f_e)}_{\text{room on } e}.$$

[for all edges

$e \in P$,

increase f_e by Δ .

T.b.e.